RABBITMQ NA REDIS

Production Administration Guide

Mwongozo wa System Administrator: deployment, security, monitoring, backup, troubleshooting na recovery

Mwongozo wa mafunzo na operational reference kwa production environments.

1. Utangulizi na Majukumu ya System Administrator

RabbitMQ ni message broker inayosambaza messages kati ya producer na consumer. Redis ni in-memory data store inayoweza kutumika kwa cache, sessions, counters, locks, queues au Celery.

Kazi ya administrator ni kuhakikisha services zinapatikana, ziko salama, zina performance nzuri, zina monitoring na zinaweza kurecover baada ya failure.

Majukumu makuu

• Availability na reliability

• Performance monitoring: CPU, RAM, disk, connections na latency

• Security: users, permissions, TLS na firewall

• Backup, restore na disaster recovery

• Maintenance, patching na upgrades

• Incident response na documentation

2. RabbitMQ: Concepts na Production Responsibilities

Flow ya kawaida ni Producer → Exchange → Binding/Routing Key → Queue → Consumer. Developer anatakiwa kutoa messaging requirements; administrator anasimamia broker na infrastructure.

Maswali ya kumuuliza developer

• Majina ya exchanges, queues na virtual hosts

• Exchange type na routing keys

• Idadi ya consumers na expected throughput

• ACK, retry na dead-letter strategy

• Mahitaji ya persistence na recovery

• Application IPs, ports, credentials na TLS requirements

3. RabbitMQ: Practical Administration

Kuthibitisha container na service health

```
docker ps --filter name=rabbitmq
docker logs --tail 100 rabbitmq
docker exec -it rabbitmq rabbitmq-diagnostics status
```

Kuchunguza queues na consumers

```
docker exec -it rabbitmq rabbitmqctl list_queues \
name messages_ready messages_unacknowledged consumers
```

Connections, users na permissions

```
docker exec -it rabbitmq rabbitmqctl list_connections
docker exec -it rabbitmq rabbitmqctl list_users
docker exec -it rabbitmq rabbitmqctl list_vhosts
docker exec -it rabbitmq rabbitmqctl list_permissions -p /
```

Ports muhimu: 5672 AMQP, 15672 Management UI/HTTP API, 5671 AMQPS. Zuia ports kwa firewall na usiweke Management UI wazi bila controls za security.

Usifute queue au kufanya purge bila kuthibitisha impact, owner wa application na recovery plan.

4. Redis: Concepts na Production Responsibilities

Redis inaweza kutumika kwa cache, sessions, Celery broker, task results, counters, distributed locks au persistent data. Use case ndiyo huamua persistence, TTL, memory policy na recovery plan.

Maswali ya msingi

• Redis inatumika kwa cache, sessions, queue, broker au data muhimu?

• Ni database index gani inatumika?

• Key TTL na eviction policy ni zipi?

• Expected memory, clients na commands per second ni kiasi gani?

• Connection pool, timeout na retry behavior ikoje?

• Ni data ipi lazima irecover?

5. Redis: Practical Administration

Health, logs na connectivity

```
docker ps --filter name=redis
docker logs --tail 100 redis
docker exec -it redis redis-cli ping
```

Memory, clients na persistence

```
docker exec -it redis redis-cli INFO memory
docker exec -it redis redis-cli INFO clients
docker exec -it redis redis-cli INFO persistence
docker exec -it redis redis-cli CONFIG GET maxmemory
```

Kuchunguza keys kwa usalama

```
docker exec -it redis redis-cli DBSIZE
docker exec -it redis redis-cli SCAN 0 COUNT 100
```

Backup snapshot

```
docker exec -it redis redis-cli BGSAVE
```

Epuka KEYS * kwenye production yenye keys nyingi; tumia SCAN. BGSAVE siyo backup kamili mpaka snapshot files zihifadhiwe kwenye destination ya backup na restore ijapimwe.

6. Monitoring na Incident Response

RabbitMQ metrics

• Queue depth na growth rate

• Publish rate dhidi ya delivery/ack rate

• Consumers na unacknowledged messages

• Memory, disk free space na node health

• Dead-letter queues na failed messages

Redis metrics

• Used memory, maxmemory na evicted keys

• Connected/blocked clients

• Command rate na latency

• Cache hit ratio na expired keys

• RDB/AOF persistence status

Incident workflow

• Thibitisha incident na impact

• Angalia host, Docker, network, CPU, RAM na disk

• Review logs na metrics

• Kusanya timestamps na recent changes

• Mitigate kwa hatua salama; usifanye purge au FLUSHALL kama hatua ya kwanza

• Escalate kwa developer na document incident

7. Backup, Restore na Disaster Recovery

RabbitMQ definitions backup inaweza kuhifadhi users, vhosts, queues, exchanges na bindings. Hii si lazima iwe backup ya messages zote; message recovery inahitaji strategy maalum.

Redis backup inategemea use case. Cache inaweza kujengwa upya, lakini sessions, queues au data muhimu inaweza kuhitaji recovery.

Backup lazima iwe nje ya container/host inapowezekana, iwe na access control, retention, encryption na restore testing.

Recovery checklist

• Tambua RPO na RTO

• Hifadhi backup kwenye destination tofauti

• Punguza access na encrypt backups

• Fanya restore drills

• Document recovery steps na owners

8. Security na Capacity Planning

Security baseline

• Usifungue Redis kwa public internet

• Zuia RabbitMQ Management UI kwa firewall/VPN

• Tumia separate users na least privilege

• Tumia TLS inapohitajika

• Usiweke credentials kwenye Git au logs

• Patch host na images kwa change-management process

Capacity planning

• Fuatilia trends za memory, disk, queues na clients

• Fanya load testing kabla ya traffic kubwa

• Ongeza consumers kwa uangalifu

• Chagua maxmemory/eviction policy kulingana na data criticality

• HA inahitaji replication, failure detection na failover iliyojaribiwa

9. Production Checklists

Kila siku

• Service/container health

• CPU, RAM na disk

• RabbitMQ queues, consumers na unacknowledged messages

• Redis memory, clients na errors

• Monitoring alerts na critical logs

Kila wiki

• Queue trends na failed tasks

• Redis evictions, latency na memory growth

• Verify backup jobs

• Review incidents

Kila mwezi

• Users, permissions na firewall

• Security updates na image versions

• Restore test

• Capacity plan na documentation

10. Developer Handover Questions

RabbitMQ

• Producer na consumer ni services zipi?

• Exchange, queue, routing key na vhost ni zipi?

• ACK, retry na dead-letter strategy ni ipi?

• Expected throughput ni kiasi gani?

• Ni data gani lazima irecover?

Redis

• Use case ni ipi?

• Database index, TTL na eviction policy ni zipi?

• Data inaweza kupotea wakati wa restart?

• Connection pool na timeout ni zipi?

• RPO/RTO ni ipi?

Hitimisho

System Administrator anasimamia availability, security, performance, persistence, monitoring na recovery. Developer anafafanua application behavior na requirements; administrator anahakikisha infrastructure inatimiza mahitaji hayo.

Tahadhari: Commands ni mifano ya kuanzia. Badilisha majina ya containers, users, paths na environment kulingana na mfumo wako. Fanya mabadiliko makubwa kwenye maintenance window na kwa approval inayofaa.